

From Human–AI Performance to Sustained Capability

Toward a Design Science of Human–AI Developmental Ecologies

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Abstract

Artificial intelligence is arriving amid a period of unusually dense and interacting global challenges: ecological disruption, geopolitical instability, institutional fragility, economic transformation, information disorder, technological acceleration, and persistent failures of collective coordination. These problems differ in their causes and mechanisms, but together they pose a recurring civilizational question:

How can human beings and human systems become more capable of responding wisely to the conditions they are creating?

The language of *polycrisis* is itself contested, but a growing systems literature examines how crises can become causally entangled through common stresses, cascading effects, and cross-system feedbacks (Lawrence et al., 2024). Resilience and adaptive-governance traditions likewise emphasize the capacity of social–ecological systems to reorganize under disturbance and the importance of interactions among people, organizations, institutions, and levels of governance (Walker et al., 2004; Folke et al., 2005).

Many traditions have addressed parts of this problem through consciousness, human development, sustainability, education, institutional reform, deliberation, intentional community, and technological design. Their limitations are not uniform, but several recurring difficulties appear. Some offer profound meaning but weak causal traction. Some generate concrete local benefit without mechanisms capable of propagating broadly. Some address development without materially changing the systems within which human capacities are exercised. Others engage civilizational problems while offering relatively weak leverage over the infrastructures increasingly organizing social life.

Artificial intelligence is distinctive because it cuts across these levels simultaneously. AI increasingly participates in cognition, learning, scientific inquiry, economic production, institutional decision-making, education, culture, communication, governance, and collective problem-solving. Economic and organizational research already treats AI as consequential not only for productivity but for institutional structure, decision-making power, information flows, human wellbeing, and transition dynamics (Brynjolfsson, Korinek, & Agrawal, 2025; Bankins et al., 2026).

It is not merely another proposal for how people should think or behave. It is becoming part of the environments through which people think, learn, decide, relate, organize, create knowledge, and exercise agency.

AI can therefore be understood not only as a new technological capability but as a historically consequential intervention into human and collective capability.

At the same time, human–AI research is itself moving beyond model-centered evaluation. Recent reviews identify interaction design, changing human–machine relationships, organizational and societal implications, multilevel effects, and the need for more longitudinal and context-sensitive research as important frontiers. They also continue to describe theoretical fragmentation and limitations in integrative evaluation (Seini, Adam, & Preko, 2026; Bankins et al., 2026). Post-deployment monitoring research similarly identifies human–AI feedback loops, beneficial-impact measurement, and validated real-world monitoring methods as unresolved methodological problems (Rao et al., 2026).

The emerging problem is therefore not that these dimensions have been overlooked. It is that recognition of them does not yet provide a cumulative account of **how configurations of artificial capability, human participation, institutions, and history interact to produce trajectories across levels and time.**

This paper develops a prospective research architecture organized around that problem. It distinguishes present performance from **sustained capability**; treats recurrent distributions of consequential operations as a possible developmental mechanism; situates local human–AI developmental ecologies inside larger organizational, institutional, societal, political-economic, and ecological contexts; and asks how outcomes propagate across microgenetic, mesogenetic, and longitudinal timescales.

The Human–AI Developmental Research program (HAIDR) is presented not as a comprehensive theory of human–AI effects, but as an integrative architecture for connecting **nested ecology, configuration, participation, mechanism, outcome, and trajectory** across human, coupled, collective, and institutional levels.

The longer-term ambition is a design science capable of asking:

Which human–AI configurations produce which capability trajectories, for whom, under what conditions, across which levels and timescales?

1. From Artificial Capability to Sociotechnical Trajectory

Contemporary AI evaluation already extends well beyond model benchmarks. Human–AI research investigates teaming, function allocation, cognitive offloading, learning and skill, organizational adoption, agentic oversight, collective effects, and post-deployment behavior. Recent

interdisciplinary reviews explicitly describe the field as multilevel and context-dependent while also finding substantial fragmentation across disciplinary and analytical levels (Seini et al., 2026; Bankins et al., 2026).

The scientific transition is therefore not adequately characterized as a move from a field concerned only with model capability toward one newly discovering human and social effects.

A more accurate transition is:

from increasingly sophisticated but partially separated forms of evaluation toward cumulative inference across configurations, levels, and timescales.

Several empirical and theoretical literatures already establish pieces of this problem.

Human–AI performance is highly heterogeneous: a preregistered meta-analysis of 106 experiments found that human–AI combinations improved on humans alone on average, but did not on average outperform the better of humans or AI alone, with outcomes varying substantially by task and relative capability (Vaccaro, Almaatouq, & Malone, 2024).

Classic automation research established much earlier that automation does not merely substitute for human activity; it reorganizes what humans do and creates new coordination and supervisory demands (Parasuraman, Sheridan, & Wickens, 2000).

Organizational reviews now similarly emphasize that the effects of AI depend on interactions among individual, group, organizational, and broader contextual factors rather than on technological capability alone (Bankins et al., 2024, 2026).

And post-deployment evaluation remains comparatively immature: NIST's 2026 review describes validated monitoring methodologies and common terminology as nascent and scattered, while specifically identifying insufficient research on human–AI feedback loops and metrics for beneficial impacts to humans (Rao et al., 2026).

We therefore increasingly know that **configuration matters**. What remains harder is connecting these observations strongly enough to answer:

Which configurations cause which durable consequences, under what conditions, through which mechanisms, and with what degree of transportability?

This difficulty is especially important because conclusions at one level do not automatically license conclusions at another.

- Improved task performance does not establish increased human competence.
- Higher individual productivity does not establish greater collective intelligence.
- Organizational throughput does not establish preserved expertise or resilience.
- Formal human accountability does not establish practical capacity to intervene.
- And successful AI applications to individual social problems do not establish increased societal adaptive capacity.

The distinction between levels is not merely theoretical. For example, recent meta-analytic evidence suggests that generative AI can improve local creative production while simultaneously producing a small but significant homogenization of outputs across participants, illustrating how individual and collective outcomes can diverge (de Rooij & Biskjaer, 2026).

The resulting frontier is therefore a problem of **sociotechnical evaluation across levels and time**.

The relevant system increasingly includes not simply the model but the human participants, interaction protocol, workflow, organization, incentives, institutions, infrastructures, and histories through which model capabilities are realized.

The central research question becomes:

How do configurations of artificial capability, human participation, interaction design, delegation, institutions, and history alter human and collective capability over time?

2. Human–AI Developmental Ecologies

I use **human–AI developmental ecology** to refer to the structured set of conditions through which recurrent human–AI participation occurs and through which its developmental consequences are produced.

This is a synthetic construct rather than a claim that the field lacks adjacent concepts.

Distributed-cognition research has long argued that cognition should sometimes be analyzed across whole environments of people, artifacts, and technologies rather than confined to the individual (Hollan, Hutchins, & Kirsh, 2000). Cognitive-offloading research likewise examines how people alter the information-processing demands of tasks through external action and technological support (Risko & Gilbert, 2016). Human-factors research analyzes distributions of function between humans and automation, while Coactive Design emphasizes interdependence rather than treating autonomy as an isolated system property (Parasuraman et al., 2000; Johnson et al., 2014).

HAIDR's developmental-ecology proposal builds on these traditions while adding a particular longitudinal question:

How does repeated participation in a given sociotechnical configuration alter what its participants subsequently become capable of doing?

Crucially, the human–AI ecology is not a closed dyad. It is **nested within larger ecologies**. A person interacting with an AI system is simultaneously situated within a task environment, social and professional setting, organization, institutional and regulatory regime, political economy, culture, technological infrastructure, and material environment.

This nesting has precedent in multilevel approaches to AI and work, which explicitly distinguish individual, group, organizational, and societal factors, and in social–ecological governance research, which analyzes adaptation across interacting organizational levels (Bankins et al., 2024, 2026; Folke et al., 2005).

The developmental ecology therefore has at least two analytical dimensions:

1. the **local configuration of participation** through which human and AI activity is distributed; and
2. the **nested environments** that select, reward, constrain, reproduce, or transform that configuration.

The same interaction protocol could plausibly have different developmental consequences in a classroom designed around learning, a workplace optimized for throughput, an emergency-response environment, or a regulated clinical setting.

This is not yet a settled empirical generalization. It is a design hypothesis implied by literatures showing that AI outcomes are contingent on organizational, task, user, and institutional conditions.

Configuration variables

Several interacting variable classes constitute the local ecology.

Artificial-system variables

These include model capability, modality, memory, personalization, continuity, tool access, planning horizon, autonomy, transparency, initiative, and capacity for persistent action.

They determine the space of technically possible participation.

Interaction variables

These include interface, timing of assistance, feedback structure, provenance visibility, modality, task decomposition, friction, and opportunities to revise.

Participation and delegation variables

These concern who:

- establishes goals;
- frames the problem;
- decomposes the task;
- initiates action;
- retrieves information;
- remembers;
- generates possibilities;
- interprets evidence;
- plans;

- evaluates;
- monitors;
- detects error;
- corrects;
- decides;
- executes;
- and determines completion.

The ancestry here is explicit. Parasuraman et al. distinguish automation of information acquisition, information analysis, decision/action selection, and action implementation; Coactive Design emphasizes observability, predictability, directability, and interdependence in joint activity (Parasuraman et al., 2000; Johnson et al., 2014). HAIDR's proposed operation-level representation extends this logic longitudinally rather than replacing it.

Authority matters as much as activity. A human may nominally review a decision while possessing little effective opportunity or competence to alter it. Recent research on software-agent oversight similarly finds that practical oversight comprises several different forms of work—including a priori control, co-planning, real-time monitoring, and post hoc review—rather than a single generic “human-in-the-loop” function (Dhanorkar, Passi, & Vorvoreanu, 2026).

Human variables

Relevant factors include prior knowledge, expertise, developmental history, metacognitive ability, goals, motivation, trust, identity, agency, existing strategy repertoire, and prior history of AI participation.

Relational and social variables

These include the perceived role of AI—tool, tutor, collaborator, coach, authority, companion, critic—as well as human peers, mentors, teams, norms, and distributed sources of knowledge.

Temporal variables

These include duration, recurrence, sequence, continuity, scaffold fading, interruptions, model changes, cumulative dependence, prior adaptations, and opportunities for independent practice.

Nested ecological levels

These local variables are embedded inside broader conditions:

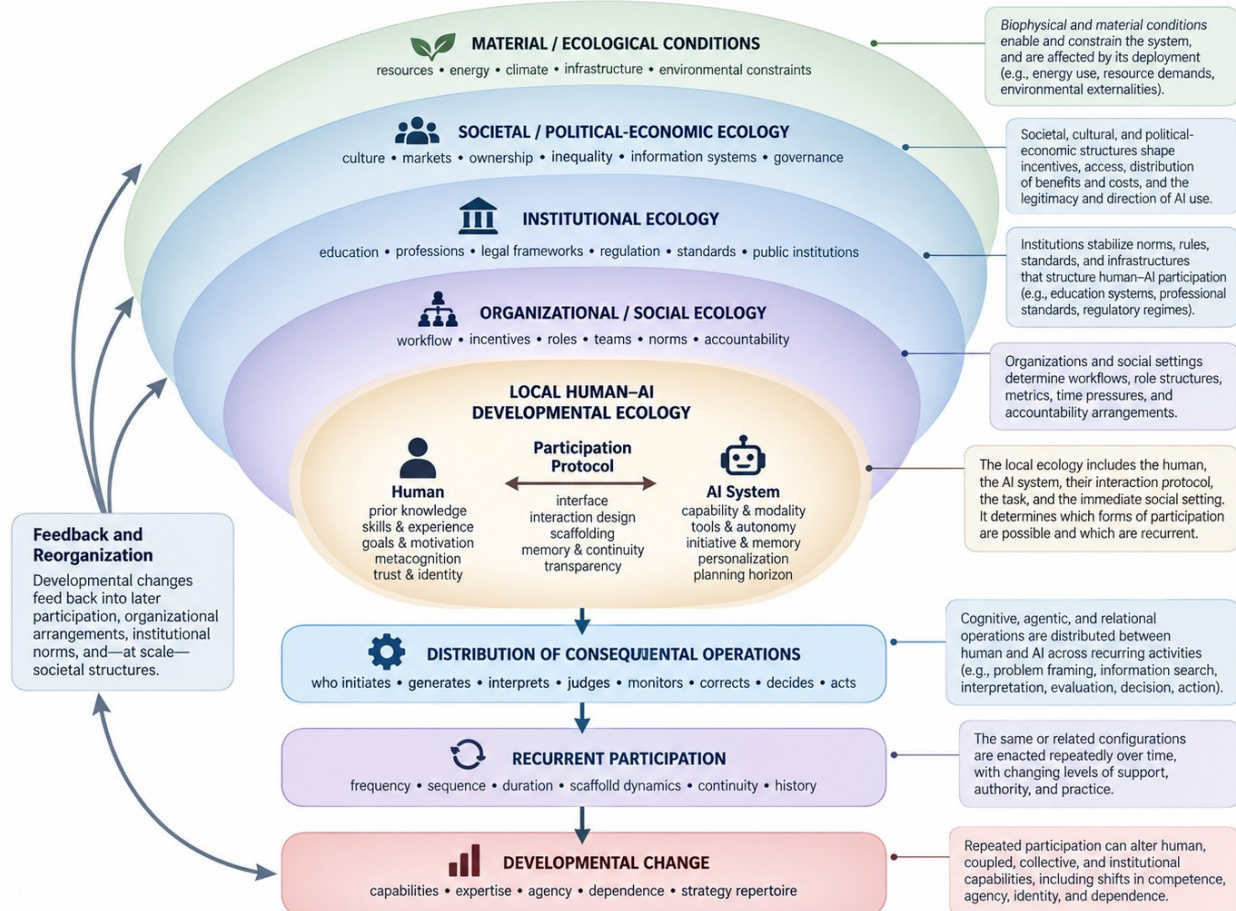
- **task ecology**—What kinds of problems, uncertainty, constraints, and stakes structure the activity?
- **organizational ecology**—What workflows, incentives, role structures, metrics, time pressures, and accountability arrangements govern participation?
- **institutional ecology**—What professions, educational systems, legal rules, regulatory regimes, standards, and infrastructures stabilize the arrangement?

- **societal and cultural ecology**—What norms, knowledge institutions, media systems, social inequalities, and patterns of participation surround it?
- **political-economic ecology**—Who owns models and infrastructure? How are benefits, costs, authority, and dependencies distributed? Economic analyses of transformative AI already identify income distribution, decision-making power, information flows, wellbeing, geoeconomics, and transition dynamics as major research problems, giving this political-economic layer a direct scholarly anchor (Brynjolfsson et al., 2025).
- **material and ecological context**—What energetic, environmental, infrastructural, and resource conditions enable the system, and what externalities does its deployment create?

The developmental ecology is therefore not only the interface immediately surrounding a user. It is the **nested configuration through which artificial capability becomes socially organized participation**.

Figure 1. Nested Human–AI Developmental Ecology

Human–AI development occurs within a nested set of ecological contexts. Local configurations of human and AI participation produce developmental change, which in turn feeds back into organizations, institutions, societies, and material conditions.



Developmental changes feed back into subsequent participation, organizational arrangements, institutional norms, and—in aggregate—societal structure.

The scientific object is therefore not simply: **What does GPT-X do to people?** —but: **What nested developmental ecology does a person or collective repeatedly participate in, and what trajectories does that ecology tend to produce?**

3. Sustained Capability

The developmental framing introduces a distinction between **present performance** and **sustained capability**.

This distinction has strong intellectual ancestry. Learning science has long distinguished observable performance during acquisition from relatively durable learning expressed through retention and transfer. Improvements in immediate performance can occur without corresponding long-term learning, and the reverse can also occur (Soderstrom & Bjork, 2015).

Automation research similarly documents cases in which high automated performance can coexist with reduced human situation awareness or poorer ability to resume control after automation failure (Bainbridge, 1983; Endsley & Kiris, 1995).

Cognitive-offloading research complicates any simplistic conclusion that externalization is inherently harmful: offloading can reduce internal cognitive demand and can be adaptive, while its consequences depend on metacognition, task requirements, and the nature of the external support (Risko & Gilbert, 2016).

HAIDR therefore does **not** equate desirable development with maximization of unaided performance.

Two human–AI arrangements can produce comparable immediate results while generating different trajectories.

- One may strengthen independent judgment.
- Another may support stable complementary specialization.
- Another may remove opportunities through which expertise would otherwise develop.
- Another may weaken lower-level fluency while strengthening higher-order supervision.
- Another may increase organizational productivity while eroding the distributed expertise required for future correction.

“Augmentation” and “replacement” are therefore often too coarse. A more useful evaluation requires a multidimensional **outcome vector** distinguishing:

- present task performance;
- supported human–AI capability;
- retained human competence;
- practical intervention capability;

- higher-order orchestration;
- collective knowledge distribution;
- organizational capability;
- resilience under perturbation;
- infrastructure dependence;
- and changes in future strategy space.

The core evidential question becomes:

What observations, comparisons, causal assumptions, and validation standards justify the conclusion that a human–AI configuration produces sustained capability rather than merely strong present performance?

Existing literatures establish parts of this problem. The proposed contribution is their explicit conjunction under persistent AI participation. Unaided performance is therefore not a universal gold standard. Stable external support may constitute valuable capability. Some operations may reasonably migrate to artificial systems. Lower-level delegation may permit humans to develop new higher-order capacities.

The question is instead:

What capability changed, where does it now reside, what does it depend upon, and what happens when conditions change?

This leads to a central integrative problem:

How does persistent redistribution of cognitive, agentic, and relational activity between humans and AI alter the future capacities and agency of the human, the capabilities of the coupled system, and the institutional conditions under which those trajectories become stabilized?

And then:

How should those divergent outcomes be evaluated and governed?

4. Participation History as a Developmental Variable

Nominal task allocation often conceals developmentally consequential differences.

“Human review,” for example, can mean:

- independent judgment before AI exposure;
- judgment after AI recommendation;
- comparison between independently generated human and AI answers;
- targeted anomaly detection;

- substantive correction;
- procedural approval;
- or near-total deference accompanied by nominal sign-off.

Automation and human-factors research provides direct precedent for treating such differences as consequential. Bainbridge's "ironies of automation" highlighted the problem of leaving humans responsible for rare abnormal conditions while routine activity—and thus routine practice—is automated. Endsley and Kiris subsequently demonstrated poorer situation awareness and takeover performance under higher automation in an experimental navigation task (Bainbridge, 1983; Endsley & Kiris, 1995).

A developmentally informative representation may therefore need to record the actual participation history of consequential operations through variables such as:

operation × actor × initiative × authority × sequence × timing of assistance × information provenance × feedback × opportunity for revision × recurrence

This particular representation is a **HAIDR proposal**, not an established taxonomy.

Several distinctions are especially important:

Initiative—Who begins the cognitive or agentic operation?

Sequence—Does human judgment occur before or after exposure to AI output?

Authority—Who can meaningfully revise, reject, override, or redirect?

Epistemic provenance—Are multiple judgments genuinely independent or generated from a common artificial source?

Generative contribution—Who generates hypotheses, distinctions, alternatives, interpretations, or plans?

Evaluative contribution—Who compares alternatives, establishes criteria, identifies error, and determines sufficiency?

Corrective contribution—Who notices failure and performs the work required to repair it?

Recurrence—Which operations are repeatedly practiced, intermittently practiced, or effectively abandoned?

Scaffold dynamics—Does support remain constant, increase, fade, or adapt to demonstrated competence?

Transfer opportunity—Does the ecology require operations to be enacted under new conditions, with reduced support, or outside the original coupling?

The proposal is not that these variables constitute an entirely new work-analysis framework.

They should be compared against established function-allocation models, human-automation research, Coactive Design, mixed-initiative systems, and workplace-learning approaches (Parasuraman et al., 2000; Johnson et al., 2014).

Participation-history measures earn their place only if they improve **discrimination, prediction, intervention, or transportability**.

This is the intended function of integrative work: not to construct another encompassing vocabulary, but to identify distinctions at the interfaces of established literatures that materially change empirical inference.

5. HAIDR as a Nested, Multiscale Research Architecture

The Human–AI Developmental Research program can therefore be stated as:

HAIDR is a candidate integrative research architecture for investigating how persistent sociotechnical configurations, situated within nested institutional and societal ecologies, shape trajectories of human, coupled, collective, and institutional capability.

A simple linear representation is insufficient.

The architecture has three interacting dimensions:

1. **nested ecological level;**
2. **developmental process;**
3. **timescale.**

Ecological embedding

At the outer levels are material/ecological, societal, political-economic, institutional, and organizational conditions.

Within them sits the local human–AI configuration.

This multilevel orientation has strong precedent in organizational AI scholarship and adaptive-governance research, although HAIDR's particular nesting and developmental interpretation are its own synthesis (Bankins et al., 2024, 2026; Folke et al., 2005).

Higher-order environments shape which local configurations become feasible, profitable, legitimate, mandatory, or normal.

Local configurations can in turn aggregate upward, altering professions, organizations, institutions, markets, norms, and infrastructures.

Developmental process

Within a given ecology:

configuration

- **distribution of consequential operations**
- **recurrent participation**
- **developmental mechanisms**
- **multilevel outcome vector**
- **trajectory**

But the pathway is recursive:

- Changed capacities alter later participation.
- Changed participation alters roles and workflows.
- Changed workflows can become institutionalized.
- Institutionalized arrangements alter subsequent individual ecologies.

Timescale

Developmental science provides useful methodological ancestry for separating timescales of change.

Siegler and Crowley's microgenetic method uses dense observations across periods of change to study the processes through which new strategies emerge (Siegler & Crowley, 1991). Michael Cole's mesogenetic approach explicitly argues that development occurs across intersecting temporal and contextual scales and calls for simultaneous attention to histories of persons and developmental contexts (Cole, 2015).

HAIDR adapts this distinction as follows.

Microgenetic—change occurring within particular episodes of participation.

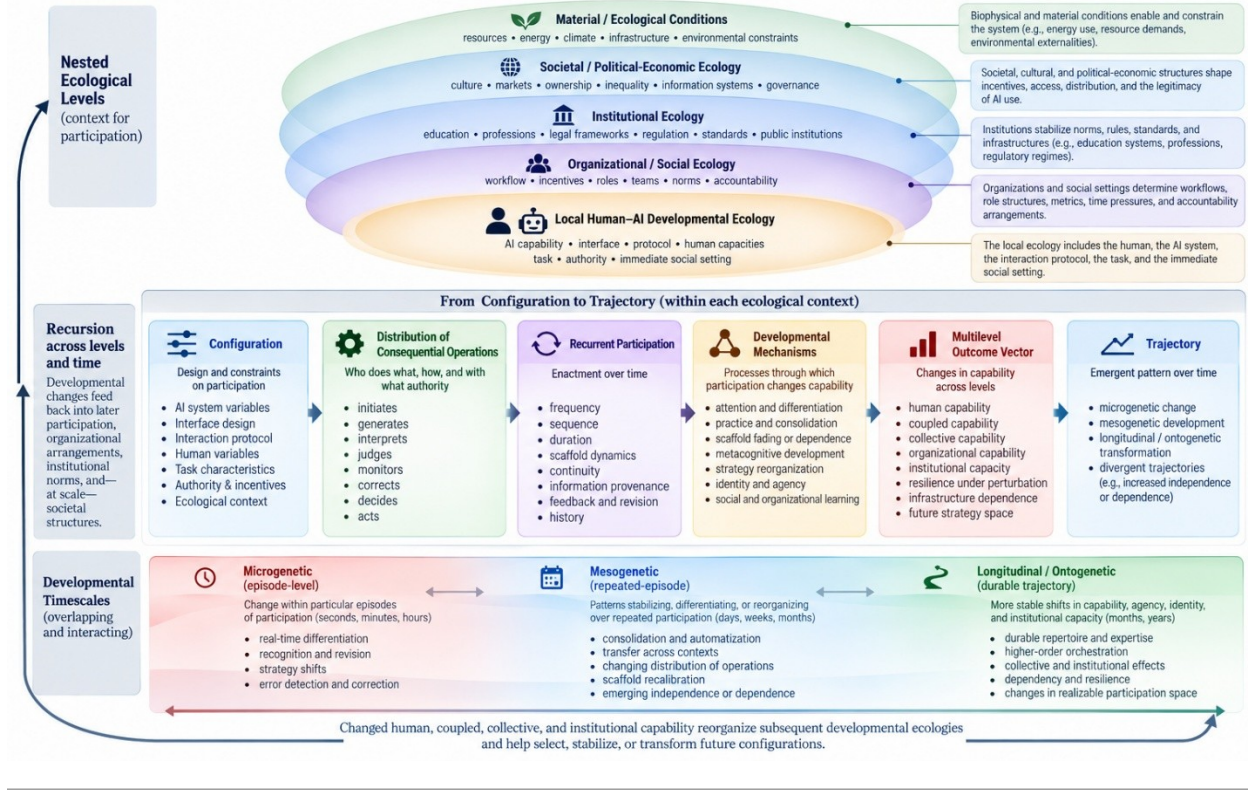
Mesogenetic—patterns stabilizing, differentiating, transferring, or reorganizing across repeated episodes over days, weeks, or months.

Longitudinal / ontogenetic—durable shifts in repertoire, agency, expertise, dependence, identity, role, and realizable participation across longer periods.

Organizational and institutional trajectories can likewise operate at longer collective timescales.

Figure 2. HAIDR Multiscale Architecture

Human–AI developmental ecologies operate across nested ecological levels and developmental timescales. Configurations of participation shape developmental mechanisms and multilevel outcomes, which recursively alter subsequent ecologies.



6. Research Sites Across Scales

HAIDR is too broad to operationalize in a single study. Its value therefore depends on bounded research sites through which different parts of the architecture can be investigated. **Current research sites include AI-Mediated Reflection, anchored empirically by Study 0; organizational systems; neuroscience and passive sensing; collective intelligence; and Wunderhorn (see below).** The broader program also includes prospective work on **developmental measurement and inference, AI-mediated research-methodology acquisition, and adaptive developmental interventions.**

6.1 AI-Mediated Reflection

AI-mediated reflection (AIMR) offers a particularly tractable microdevelopmental site because the relevant operations—attention, interpretation, differentiation, metacognition, conceptualization, self-questioning, and meaning-making—can be observed directly in dialogue and then tested for persistence and transfer.

AIMR sits adjacent to several active literatures rather than constituting an unoccupied field. The Reflective Agency framework, for example, explicitly examines how AI-mediated reflective systems may shape rather than merely support self-understanding and identifies tensions involving internal

origination, automation, ambiguity, mediation, and autonomy (Kim et al., 2025). Systematic reviews of generative AI and self-regulated learning similarly identify goal-setting, monitoring, strategy selection, feedback, and self-reflection as important loci of AI mediation while noting uncertainty about underlying mechanisms (Xia et al., 2026).

The primary AIMR question is therefore not:

Can AI produce useful reflections?

It is:

What forms of AI participation help a person become a better reflector rather than merely giving the person better reflections?

The distinction is between **better reflection while using AI** and **greater reflective capability because of repeated participation in a particular human–AI reflective ecology**.

This is a developmental extension of existing reflective-agency and self-regulation questions, not a claim that AI-mediated reflection itself is novel.

AIMR Across Developmental Timescales

A single reflective exchange can produce an insight without producing development.

AIMR therefore distinguishes at least three nested timescales:

Microgenetic: transformation within an episode

Within an individual reflective episode, the sequence might involve:

- experience or unresolved problem**
- **initial human formulation**
- **AI probe, question, counterexample, or interpretation**
- **differentiation**
- **recognition**
- **revision / integration**
- **provisional conclusion or action**

The microgenetic object is not simply the quality of the final answer. It is **how reflective organization changes through the encounter**.

The methodological logic follows the broader microgenetic tradition: densely observe change as it happens rather than infer developmental process only from widely separated before-and-after measurements (Siegler & Crowley, 1991).

- Which distinction was generated by whom?
- What challenge produced reorganization?
- Was the user adopting an AI interpretation or discriminating among alternatives?
- Did the exchange create a strategy likely to become available later?

The Recognition-Cycle Subloop

Recognition warrants special treatment because an externally supplied distinction does not become a human capacity merely by appearing in dialogue.

A candidate recognition cycle is:

- candidate distinction / pattern**
- **attention: “this may matter”**
- **discrimination: what exactly is being distinguished?**
- **comparison with experience / evidence**
- **accept • reject • refine**
- **reformulation in the person's own conceptual organization**
- **application to the immediate case**
- **later spontaneous recognition under altered conditions**

This cycle is a **HAIDR/AIMR hypothesis**, not a construct imported wholesale from an established recognition literature.

The last step is crucial. Recognition becomes developmentally more consequential when a distinction that was initially scaffolded becomes **independently discriminable and reusable**.

The cycle can also fail. The user may adopt fluent language without discrimination. AI may label a pattern prematurely. A compelling interpretation may be accepted because of authority or resonance rather than evidence. Recognition may remain context-bound and fail to transfer.

The subloop therefore provides a location for investigating the difference between **AI-supplied conceptual content** and **humanly available conceptual capacity**.

Figure 3. AIMR Microgenetic Loop with Recognition Subcycle

Within a single episode of AI-mediated reflection, human and AI participation can interact to produce differentiation, recognition, and integration. The recognition subcycle specifies how a candidate distinction becomes (or fails to become) a humanly available capacity.



Mesogenetic: development across repeated cycles

Across days, weeks, or months, repeated microgenetic episodes can produce a different class of effects.

This intermediate scale is consistent with Cole's mesogenetic emphasis on studying activity and person–context co-development across timescales between individual episodes and broad ontogenetic change (Cole, 2015).

The relevant questions become:

- Do distinctions recur spontaneously?
- Does the person formulate more precisely before AI intervention?
- Does AI need to supply fewer organizing concepts?
- Do strategies transfer across subject matter?
- Does the person increasingly generate counterexamples independently?
- Does ambiguity tolerance increase?
- Does the user detect when an AI interpretation is inadequate?
- Does scaffolding naturally fade?
- Does dependence increase instead?
- Does the distribution of reflective work shift?

Development might appear as increased differentiation, more independent recognition, improved metacognitive discrimination, an expanding reflective strategy repertoire, decreased need for interpretive scaffolding, or higher-order orchestration of AI itself.

Repeated participation could also produce the opposite trajectory: habitual deference, externalization of interpretation, dependence on AI-generated conceptualization, or reduced initiation.

Those alternatives make AIMR experimentally interesting precisely because current self-regulated-learning research suggests AI can support metacognitive and reflective processes while leaving mechanisms, design conditions, and risks of dependence incompletely resolved (Xia et al., 2026).

Ontogenetic: Longitudinal trajectory

At longer timescales the question becomes whether mesogenetic changes accumulate into durable transformations in reflective agency.

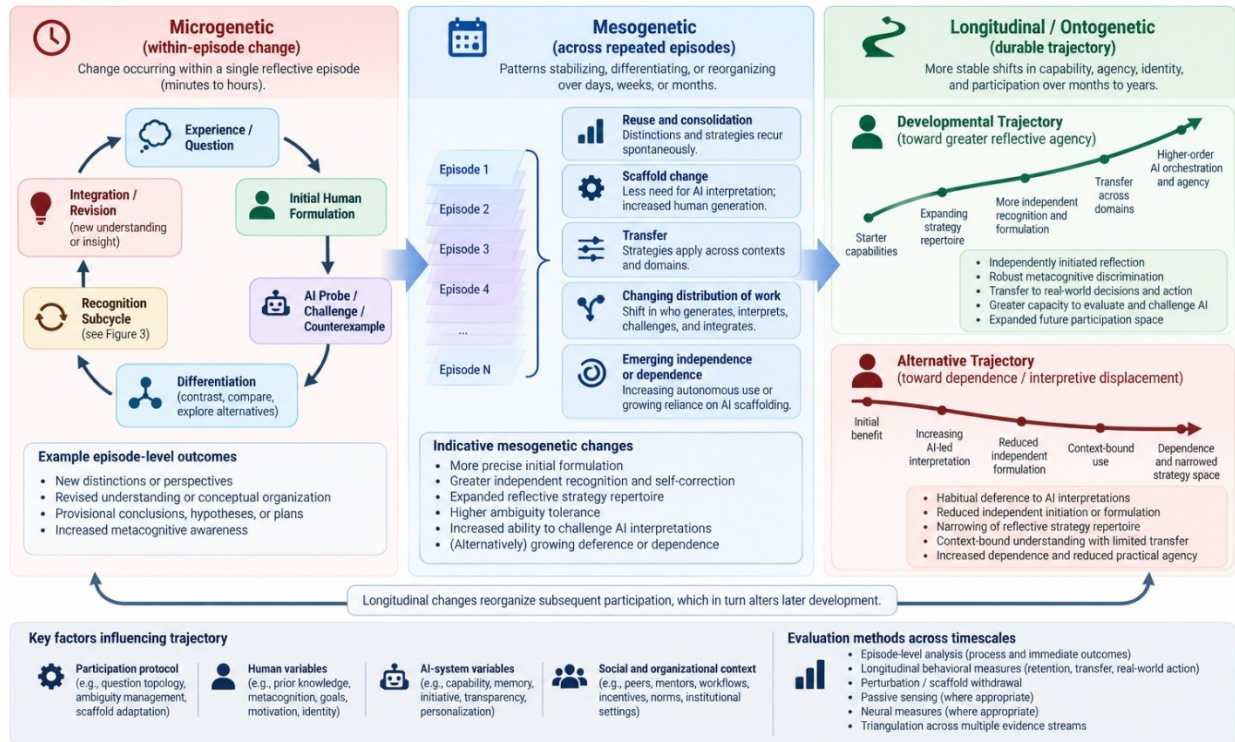
Potential outcomes include:

- independently initiated reflection;
- increased metacognitive discrimination;
- broader conceptual repertoire;
- more reliable self-correction;
- transfer across domains;
- changes in authorship of meaning;
- improved capacity to challenge externally supplied interpretations;
- higher-order coordination of AI assistance;
- altered dependence on AI;
- changes in real-world decision and action;
- and changes in future realizable participation space.

The longitudinal object is therefore not simply repeated use. It is the **changing relationship between the person, the reflective operations, and the artificial scaffold.**

Figure 4. AIMR Developmental Trajectory

Repeated cycles of AI-mediated reflection can produce different developmental trajectories. Microgenetic changes within episodes can accumulate into mesogenetic patterns and, over longer periods, more durable transformations in reflective agency.



The key empirical problem is determining which immediate interaction patterns predict which longer trajectories.

AI-Mediated Reflection Participation Protocol

The participation protocol is not simply a collection of preferred conversational behaviors.

It is the **structured distribution of reflective operations, initiative, epistemic authority, and scaffolding across time**.

Existing reflective-agency and self-regulated-learning research gives several dimensions independent scholarly grounding, while the complete protocol architecture below is an AIMR proposal (Kim et al., 2025; Xia et al., 2026).

A protocol can be represented along several dimensions.

- 1. Problem ownership**—Who identifies the problem worthy of reflection?
- 2. Initial formulation**—Does the person articulate a view before receiving AI interpretation?
- 3. Generative responsibility**—Who generates hypotheses, meanings, possibilities, interpretations, or explanatory frames?
- 4. Question topology**—Does AI ask:
 - clarifying questions;

- discriminating questions;
- evidence-seeking questions;
- counterfactual questions;
- perspective-shifting questions;
- developmental questions;
- or leading questions that implicitly supply an interpretation?

5. Timing of conceptual introduction—Does AI name a construct before the user has differentiated the phenomenon, or afterward?

6. Ambiguity management—Does the system resolve ambiguity rapidly, or preserve uncertainty long enough for the user to discriminate among possibilities?

This dimension has a particularly close neighbor in Kim et al.'s principle of *Reflective Ambiguity*, which argues for preserving interpretive space in AI-mediated self-reflection (Kim et al., 2025).

7. Challenge structure—Does AI affirm, critique, generate counterexamples, identify contradictions, expose missing evidence, or model rival interpretations?

8. Epistemic authority—Is AI treated as interpreter, hypothesis generator, critic, collaborator, or source of candidate perspectives?

9. Human revision requirement—Must the person actively reformulate after challenge, or can AI produce the synthesis?

10. Recognition requirement—Does the system merely name patterns, or require the user to identify evidence for the pattern, differentiate it from alternatives, and restate it in their own terms?

11. Alternative generation—Does the user generate competing interpretations before or after AI assistance?

12. Evidence and grounding—Does reflection remain internally coherent, or is it tested against behavior, external events, documents, other people, or observable outcomes?

13. Action coupling—Does reflection generate experiments, decisions, conversations, practices, or real-world tests capable of producing new evidence?

14. Continuity and memory—What does AI remember across episodes? Persistent memory can support developmental continuity but may also shift interpretive authority toward the system.

15. Scaffold adaptation—Does assistance remain fixed, or does the system adaptively reduce interpretation and increase demands for user generation?

16. Withdrawal—Does the protocol include intervals without AI assistance?

17. Transfer testing—Are strategies tested on novel problems, outside reflective dialogue, or after delay?

18. Meta-reflection on the coupling—Does the person periodically examine how the AI itself is shaping interpretation, agency, dependence, and vocabulary?

These dimensions create a richer protocol space than a binary distinction between AI-led and human-led reflection.

Simplified categories can nevertheless serve as experimental contrasts.

AI-led protocol:

AI identifies themes, generates interpretations, reframes, synthesizes, and recommends.

Dialogic protocol:

Human formulation precedes AI intervention; AI primarily probes, differentiates, challenges, and requests evidence.

Developmental protocol:

Scaffolding is deliberately adapted across time, with increasing demands for human generation, recognition, revision, independent enactment, and transfer.

The experimental question becomes:

Which distributions of reflective work produce strong coupled reflection while also cultivating durable human reflective agency?

Current Empirical Anchor: Study 0

Study 0 is an intensive longitudinal case of sustained human–AI reflective participation. Its retrospective phase reconstructs participation history and candidate developmental changes from the existing archive; its prospective phase uses repeated reasoning probes, delayed transfer, perturbation, and scaffold withdrawal to distinguish supported performance from persistent human capacity. Study 0 is treated as hypothesis-generating rather than as evidence sufficient to validate HAIDR's broader architecture.

6.2 Organizational Systems

Organizations provide a meso-level site for studying expertise, delegation, and practical control.

AI changes not only what work can be completed but how work is divided, how expertise is exercised, and how organizations are structured around technological capability. Reviews of AI at work emphasize precisely these multilevel interactions and the contingent nature of outcomes (Bankins et al., 2024, 2026).

Classic automation research provides a specific warning: when humans cease routinely performing operations yet remain responsible for rare failures, their capacity to intervene can degrade (Bainbridge, 1983; Endsley & Kiris, 1995).

A workflow can therefore increase immediate throughput while reducing opportunities for:

- apprenticeship;
- error detection;
- diagnostic reasoning;
- mentoring;
- independent judgment;
- and recovery when automation fails.

Alternatively, delegation may enable humans to develop higher-order coordination, evaluation, or problem-formulation capacities.

The relevant question becomes:

How do organizational and institutional arrangements stabilize particular distributions of human and artificial activity, and what capability trajectories result?

This also reveals why local interface design cannot be studied independently of institutional context. The same interaction design may plausibly have different effects under protected learning conditions than under intense production pressure. That proposition requires direct empirical testing.

6.3 Neuroscience, Passive Sensing, and Mechanistic Triangulation

Neuroscience offers a potential source of mechanistic evidence rather than a standalone criterion of development.

The candidate neuroscientific object is:

neural adaptation to recurrent distributions of cognitive operations within human–AI developmental ecologies.

The corresponding question is:

Does repeated participation in different human–AI ecologies alter the neural organization by which a person subsequently performs cognitive operations?

Direct evidence here remains preliminary.

One recent preprint combined repeated LLM-assisted essay writing, behavioral measures, and EEG across several sessions, demonstrating that longitudinal neural/behavioral designs of this general kind are feasible; it should be treated as an early methodological exemplar rather than evidence for a general neural effect of AI use (Kosmyna et al., 2025).

The strongest designs would combine several evidence streams:

Behavioral evidence

Longitudinal measures of performance, strategy use, error detection, transfer, spontaneous initiation, correction, and independent enactment.

Perturbation evidence

Performance under AI withdrawal, reduced support, changed models, conflicting information, or novel tasks.

Passive and naturalistic sensing

Digital-sensing research demonstrates the feasibility of scalable longitudinal collection through smartphones and wearables while also documenting substantial problems involving standardization, statistical power, external validity, interpretability, and methodological heterogeneity (Akre et al., 2024; Farrahi et al., 2024).

Where ethically and methodologically appropriate, passive traces could provide longitudinal evidence about how participation changes outside laboratory snapshots.

Potential sources might include:

- interaction timing and frequency;
- task-switching patterns;
- revision behavior;
- information-seeking sequences;
- linguistic or conceptual reuse;
- device-level behavioral traces;
- wearable or physiological measures;
- and communication rhythms.

Their role should be **triangulatory rather than diagnostic**.

Evidence from digital-phenotyping research shows why caution is necessary: passive measures can reveal meaningful behavioral patterns, but generalizability, missing data, context dependence, and incremental predictive validity remain persistent limitations (Leaning et al., 2024; Akre et al., 2024).

Neural evidence

Repeated fMRI, EEG, or other appropriate measurements could test whether changes in behavioral strategy are accompanied by changes in neural recruitment or coordination during independently performed tasks.

The strongest inference would therefore come from convergence:

participation history + passive longitudinal traces + behavioral change + transfer + perturbation / withdrawal + neural change

—rather than from any one channel.

Neural adaptation is not itself evidence of beneficial development. Passive sensing is not itself evidence of wellbeing or capability. But convergent longitudinal measurement may help distinguish **coupling-dependent performance, durable human change, reorganized cognitive strategy, and behavioral transfer.**

6.4 Collective Intelligence

Individual augmentation does not guarantee collective improvement.

Collective-intelligence research provides longstanding precedent for treating group capability as analytically distinct from the abilities of individual members, even though specific models of a general group “c factor” remain debated (Woolley et al., 2010).

Generative AI adds a new complication. Recent meta-analytic evidence finds a small but significant homogenization effect across AI-assisted creative outputs, despite possible improvements in individual creative performance, and finds substantial variation by task and interaction structure (de Rooij & Biskjaer, 2026).

This illustrates a broader HAIDR concern:

A system may improve **local performance** while altering the **distribution of strategies, judgments, errors, and expertise** at the collective level.

AI may:

- reduce diversity;
- increase correlated errors;
- centralize epistemic dependence;
- or alter information aggregation.

Other configurations may:

- improve access to knowledge;
- support deliberation;
- diversify hypothesis generation;
- or enable new forms of coordination.

The developmental question therefore scales from:

What is this person becoming capable of?

to:

What is this group or institution becoming capable of, and what forms of independent expertise, epistemic diversity, correction, and adaptive capacity remain distributed within it?

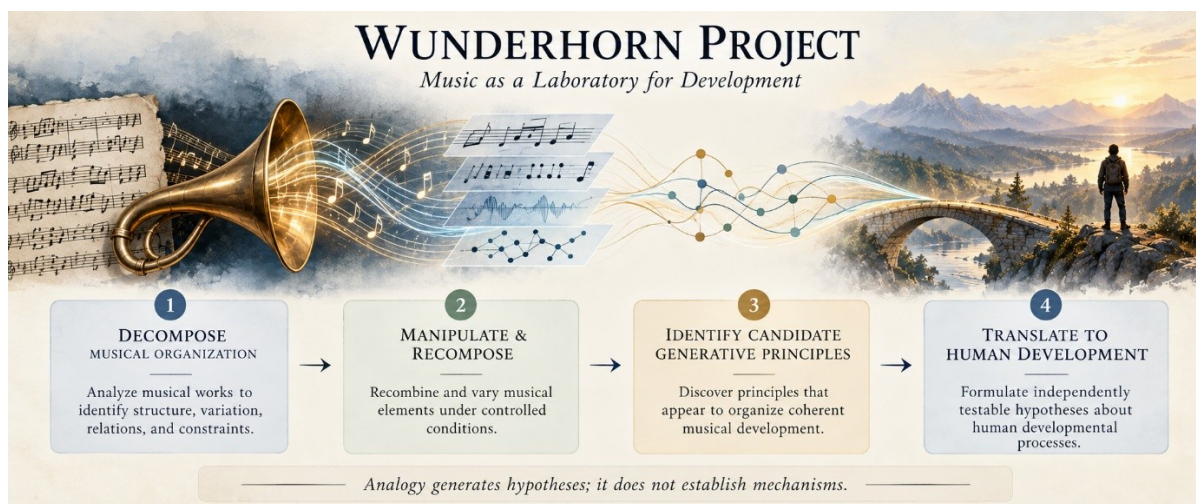
If many individuals rely on similar AI systems, apparently independent human judgments may become correlated.

If organizations stop cultivating expertise because AI performs routine cognitive operations, short-term efficiency may alter future collective resilience.

These are hypotheses requiring direct cross-level study rather than conclusions that can be inferred from existing individual-level experiments.

The relevant outcome is the evolving **architecture of collective capability**.

6.5. Wunderhorn: A Hypothesis Generator for Design Science



Wunderhorn occupies a different position in the architecture.

It uses music as a structured domain in which organization, differentiation, tension, integration, perturbation, and transformation can potentially be studied with high resolution.

Its purpose is **not** to demonstrate that musical organization and human development instantiate a common universal mechanism.

Its possible contribution is methodological:

Can a tractable domain of organized transformation generate relational hypotheses that become scientifically useful when independently operationalized elsewhere?

The sequence is:

musical organization

→ **controlled perturbation / transformation**

→ **candidate relational principle**

→ **cautious abstraction**

- **independent hypothesis**
- **independent test**
- **retain / refine / reject**

The epistemic boundary is critical:

Analogy generates hypotheses; it does not establish mechanisms.

This section intentionally requires relatively little external citation because *Wunderhorn* is a proposed research program rather than an established literature. Its neighboring intellectual traditions include resilience and adaptive-systems work concerned with disturbance, reorganization, and transformation (Walker et al., 2004; Folke et al., 2005), but those literatures do not validate cross-domain transfer from music.

Wunderhorn becomes valuable only if discoveries in the musical domain produce new discriminations, variables, or intervention hypotheses that later demonstrate incremental explanatory or predictive value in independent domains.

7. Coherence, Corrigibility, and Productive Reorganization

A candidate concept linking several parts of the program is **coherence**, although its theoretical status remains deliberately provisional.

The motivating problem is:

How should we distinguish systems that function effectively under present conditions from systems capable of productive reorganization when conditions, evidence, demands, or supports change?

This question has close conceptual neighbors in resilience research. Walker et al. distinguish resilience, adaptability, and transformability and define resilience partly in terms of the capacity to absorb disturbance and reorganize while undergoing change. Folke et al. similarly analyze adaptive governance during periods of abrupt change and emphasize learning, reorganization, multilevel institutions, and distributed knowledge (Walker et al., 2004; Folke et al., 2005).

HAIDR's **coherence** construct is not equivalent to those concepts.

- Present organization is insufficient.
- A student can perform well under heavy scaffolding while failing after withdrawal.
- An organization can optimize an AI-mediated workflow while losing expertise needed to recognize failure.
- A group can become tightly coordinated while losing epistemic diversity.

- A conceptual framework can become internally consistent while becoming less corrigible.

Coherence therefore cannot simply mean order, stability, internal consistency, or performance.

A provisional definition is:

Coherence is an evaluated quality of organization within a specified system, domain, task, timescale, and evaluative context.

Its developmental significance may depend on several distinguishable properties.

Organization

How are components differentiated and related under current conditions?

Corrigibility

Can the organization change appropriately when confronted with consequential evidence, error, conflict, or altered conditions? Corrigibility is not mere flexibility. A system may change frequently without becoming more responsive to reality.

Carry-forward

What remains after reorganization? Does the encounter generate strategies, distinctions, capacities, relationships, or structural changes that become available in future participation?

Component contribution

Where does capability reside? A coupled system may improve overall while particular human capacities strengthen, weaken, specialize, or become dependent.

Perturbational adequacy

How does the organization respond when familiar support or conditions change? Stable conditions can conceal brittleness.

The methodological importance of perturbation also has precedent in learning and automation research, where delayed testing, transfer, support changes, or automation failure reveal properties that current performance can conceal (Soderstrom & Bjork, 2015; Endsley & Kiris, 1995).

The relevant evidential object becomes:

What reorganizes, what persists, what transfers, what becomes newly possible, and what becomes dependent on continuing support?

The strongest warranted claim is not that coherence constitutes a universal developmental law.

It is that **organization, corrigibility, carry-forward, component contribution, and perturbational response may provide useful distinctions that present-performance measures obscure.**

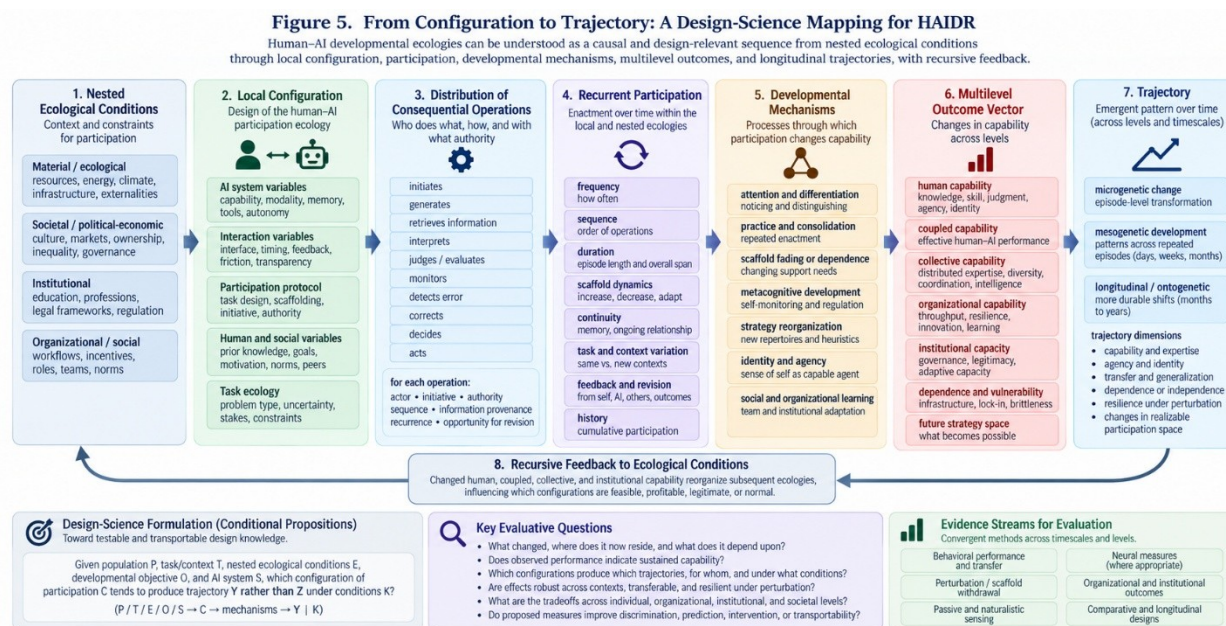
Their scientific status depends on whether they improve explanation, prediction, or intervention relative to established alternatives.

8. Toward a Design Science

The longer-term ambition is to move from configurational awareness toward **conditional design knowledge**.

Design-science traditions provide an important precedent here. Denyer, Tranfield, and van Aken propose CIMO logic, in which design propositions specify a **Context**, an **Intervention**, the **Mechanisms** through which it works, and the resulting **Outcomes**. Their purpose is precisely to move from fragmented descriptive scholarship toward conditional, solution-oriented propositions rather than universal prescriptions (Denyer, Tranfield, & van Aken, 2008).

HAIDR's mapping problem is related but adds an explicitly developmental and multilevel temporal structure:



A mature design science would eventually seek propositions of approximately the form:

Given population P, task/context T, nested ecological conditions E, developmental objective O, and AI system S, which configuration of participation C tends to produce trajectory Y rather than Z under conditions K?

This is more demanding than universal prescriptions such as: “preserve productive friction,” “keep a human in the loop,” or “augment rather than replace.”

A meaningful design proposition must specify:

- for whom;
- doing what;
- under which social and institutional conditions;
- with which distribution of operations;
- with respect to which outcome vector;
- and across what timescale.

The causal-evaluation literature adds another constraint: design claims should ultimately be supported in meaningful deployment contexts rather than inferred solely from benchmark or short-session performance. NIST's monitoring work and recent causal-field-evaluation proposals both point toward this need for post-deployment and real-world evidence (Rao et al., 2026).

A further methodological problem is how to discover which configurations are worth testing when participation space is too large to specify exhaustively through theory-driven experimentation alone. As sufficiently rich longitudinal human–AI datasets become available, increasingly capable frontier AI systems may permit naturalistic interaction archives to be transformed into structured participation-topology data. This prospective methodology—computational developmental ecology mining—would reconstruct who formulates goals, initiates, generates alternatives, plans, executes, monitors, corrects, challenges or resists, and synthesizes, together with explanation and scaffolding, friction, continuity and memory, user expertise, and task characteristics. It extends the participation-history representation developed in Section 4 into a possible discovery surface; it is not an already validated HAIDR method.

Candidate configurations could be related to later strategy choice, independent task performance, transfer, metacognitive calibration, conceptual differentiation, problem-solving repertoire, and agency/autonomy, with passive-sensing or neurobehavioral measures where appropriate in future work. The orienting question is: Which recurrent distributions of cognitive operations predict unusually large subsequent increases in independently demonstrated capability? AI-assisted coding, representation learning, statistical analysis, and appropriate causal-discovery methods could generate candidate patterns, but their representations would themselves require construct validation, reliability assessment, and comparison with simpler alternative measures. Agreement among automated coders would not establish construct validity; inferred developmental changes would require corroboration through independently demonstrated capability. Computational analysis would thus support hypothesis generation, not provide a black-box measure of development.

Large observational datasets may increase configuration-discovery power without establishing developmental causality. Candidate associations must become explicit causal hypotheses tested through prospective manipulation of participation structure—for example, initiative, planning, execution, monitoring, correction, scaffolding, friction, or persistence and memory—while relevant task and system characteristics are held approximately constant where possible. Longitudinal tests should examine delayed independent performance, transfer, perturbation response, scaffold withdrawal, strategy-space expansion, metacognitive change, and agency/autonomy. Supported

performance ≠ development: neither strong coupled output nor an observational association establishes durable human change. Causal-discovery algorithms cannot substitute for experimental identification, and transfer and withdrawal evidence remain necessary discriminations when the claim concerns independently available capability.

Once credible causal mechanisms have been established, configurational research could progress toward systematic configuration optimization. Factorial and sequential experimentation, adaptive trials, contextual bandits, Bayesian optimization, and heterogeneous treatment-effect estimation are potential methods, rather than a presently mature HAIDR toolkit. The mature design question is: Given person P, task T, developmental objective D, and context C, what participation configuration maximizes durable developmental gain? This question remains conditional on the multilevel outcomes, ecological conditions, and tradeoffs specified above. One programmatic contribution of HAIDR may therefore be a configuration ontology + developmental outcome architecture + hypothesis-generation system that identifies which contrasts are scientifically worth manipulating, without needing to conduct every large-scale experiment itself. Such a contribution requires competence in causal inference, longitudinal design, measurement, experimentation, human–AI interaction, and developmental theory. The sequence—large-scale trace discovery → discriminating causal hypotheses → longitudinal experimental tests of durable change → configuration optimization—connects theoretical differentiation and computational discovery to cumulative empirical design knowledge; conceptual architecture does not substitute for validation.

The aim is therefore not to identify one ideal relationship between humans and AI. It is to build progressively stronger knowledge about:

which configurations produce which trajectories, for whom, under what conditions, across which levels and timescales, and with which tradeoffs.

9. The Civilizational Horizon: Collective Adaptive Capacity

The empirical program begins with bounded human–AI configurations, but the motivating horizon concerns the relationship between **human capacity and civilizational possibility**.

AI is becoming part of the infrastructure through which societies conduct science, education, administration, economic activity, deliberation, forecasting, communication, and coordination.

It may strengthen capacities required to address interacting systemic challenges.

It may also generate new forms of dependency, concentration, epistemic monoculture, inequality, information instability, environmental burden, and correlated vulnerability.

The first part of this claim is well grounded in existing research; the second comprises a set of plausible and partially evidenced risks rather than one established net effect.

The broader intellectual ancestry comes from three literatures:

- **polycrisis/systemic-risk research**, which studies causal interactions among global crises (Lawrence et al., 2024);
- **resilience and adaptive governance**, which studies capacities for learning, reorganization, and transformation across institutional levels (Walker et al., 2004; Folke et al., 2005);
- and **collective intelligence**, which treats group-level cognitive capability as distinct from the performance of isolated individuals (Woolley et al., 2010).

HAIDR proposes to bring these into relation with human–AI development.

AI participates in two simultaneous pathways:

systemic challenges

→ **AI-enabled response capacity**

→ **altered resilience**

and

AI diffusion

→ **altered economic / social / institutional / ecological / geopolitical conditions**

→ **altered systemic challenges**

The latter pathway has direct economic and institutional precedent: current AI research agendas already identify decision-making power, income distribution, geoeconomics, information flows, wellbeing, and transition dynamics as major effects requiring analysis (Brynjolfsson et al., 2025).

The relevant question becomes:

Under what sociotechnical and institutional configurations does AI increase collective adaptive capacity—and under what configurations does it deepen fragility?

Relevant dimensions include:

- epistemic capacity;
- collective intelligence;
- institutional competence;
- coordination;
- distributed expertise;
- human agency;
- resilience;
- distribution of capability and power;
- and capacity for revision under changing conditions.

The deepest form of the question is:

When does greater artificial intelligence produce greater civilization-level adaptive intelligence?

This is not itself an established construct or a single empirical study. It is the societal and normative horizon within which the more bounded research program acquires significance.

10. Position and Contribution

The architecture can now be summarized more precisely:

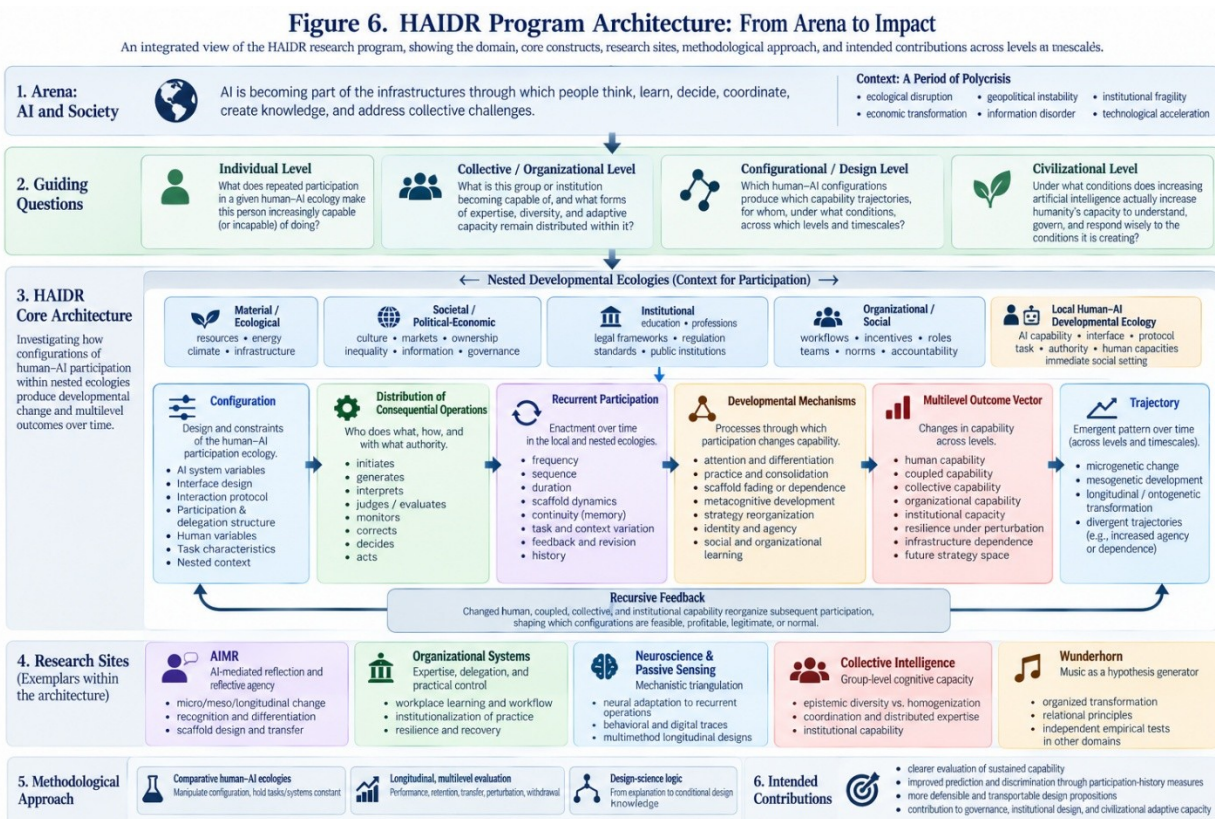


Figure 6. AI and society are the arena. Human-AI systems are the primary research territory. Nested human-AI developmental ecologies are the configurational object. Persistent distributions of consequential operations provide a candidate link between configuration and developmental change. Microgenetic → mesogenetic → longitudinal change provides the temporal architecture. Sustained capability is the recurring evaluative problem. HAIDR is the candidate integrative architecture. AIMR, organizational systems, neuroscience/passive sensing, and collective intelligence are bounded research sites. Wunderhorn is a potential generator of independently testable relational hypotheses. Coherence, corrigibility, carry-forward, component contribution, and perturbational adequacy are provisional organizing distinctions. Collective adaptive capacity is the societal horizon.

These levels should not be collapsed.

- A reflective-dialogue experiment cannot establish institutional resilience.
- Neural adaptation cannot determine which capacities ought to be preserved.
- Passive sensing cannot independently establish wellbeing or development.
- Musical structure cannot validate a human-developmental mechanism.
- Individual performance does not establish collective intelligence.
- Civilizational urgency does not substitute for causal evidence.

The prospective contribution is therefore narrower:

There appears to be a coherent unresolved problem that gains analytical leverage when these literatures are brought into explicit relation.

That proposition is consistent with recent reviews that independently describe fragmented theoretical landscapes, multilevel contingencies, and unresolved integration problems in human–AI collaboration and AI at work (Seini et al., 2026; Bankins et al., 2026).

HAIDR proposes three primary contributions for investigation.

First, clarify the evidential difference between current human–AI performance and sustained capability across human, coupled, collective, and institutional levels.

Second, test whether representing the actual distribution and history of consequential operations improves prediction and discrimination beyond established approaches to task and function allocation.

Third, develop comparative evaluation methods capable of supporting more defensible and transportable design claims about sustained human–AI arrangements.

These proposals are falsifiable:

- If existing frameworks already perform these functions adequately, additional architecture is unnecessary.
- If participation-history variables do not improve prediction, discrimination, intervention, or transport, they should be discarded.
- If developmental ecology merely redescribes existing sociotechnical constructs without additional leverage, it should carry no distinctive theoretical weight.
- If coherence contributes no incremental explanatory value, it should remain a heuristic.
- If Wunderhorn-generated hypotheses fail independent tests, they should be rejected.

The program succeeds not by encompassing human–AI research, but by helping establish **better warranted relations among ecology, configuration, participation, capability, and trajectory.**

Conclusion

Artificial intelligence is becoming part of the environments through which human beings and human systems perceive, interpret, learn, decide, coordinate, and act.

Human–AI research increasingly recognizes that its consequences depend upon task allocation, interaction design, organizational context, post-deployment behavior, and multilevel effects. Recent reviews nevertheless continue to describe substantial fragmentation, while post-deployment monitoring and longitudinal evaluation remain comparatively immature (Seini et al., 2026; Bankins et al., 2026; Rao et al., 2026).

The next challenge is to connect these observations into stronger explanations of **which arrangements produce which durable trajectories**.

The central developmental question is:

What does repeated participation in this configuration make its participants increasingly capable—or incapable—of doing later?

The corresponding integrative question is:

How does persistent redistribution of cognitive, agentic, and relational activity between humans and AI alter human capacity and agency, coupled-system capability, collective intelligence, and the nested institutions through which those trajectories become stabilized?

The design question is:

Which manipulable features of human–AI developmental ecologies produce which trajectories, for whom, under what conditions, across which levels and timescales?

And the wider civilizational question remains:

Under what conditions does increasing artificial intelligence actually increase humanity's capacity to understand, govern, and respond wisely to the conditions it is creating?

A mature science capable of answering these questions does not yet exist. **HAIDR is a proposal for helping make those questions empirically cumulative, developmentally explicit, and eventually designable.**

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